Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2023

Course: B.Tech | ***Time: 03 Hours***

Code: 104301 | ***Subject: Basic Electronics*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.

*(ii) There are **NINE** questions in this paper.*

*(iii) Attempt **FIVE** questions in all.*

(iv) Question No. 1 is compulsory.

Q.1 ***Answer the following questions (any seven only):*** **[2 x 7 = 14]**

(a) Define Doping.
(b) What are the two types of extrinsic semiconductors?
(c) Write any two differences between Zener breakdown and Avalanche breakdown.
(d) What materials are used to construct an LED?
(e) What is transistor? Give its circuit symbol.
(f) Give the relation between α and β.
(g) Why base made thin in BJT?
(h) What are the applications of JFET?
(i) What are the differences between JFET and MOSFET?
(j) Write Shockley's equation.

Q.2 (a) With a neat diagrams explain the working of a PN junction diode in forward bias and reverse bias and plot the V-I characteristics. **[7]**

(b) Explain the switching characteristic of PN junction diode. **[7]**

Q.3 (a) Draw and Explain V-I characteristics of Zener diode. **[7]**

(b) With neat diagram explain the construction and working of LED. **[7]**

Q.4 (a) With a neat sketch explain the operation of CLC (ie π section) and LC filter. **[7]**

(b) Describe the working principle of full wave rectifier with centre tapped transformer and derive the expressions for the ripple factor and efficiency. **[7]**

Q.5 (a) Explain the working of NPN and PNP transistor. **[7]**

(b) Draw and explain the input and output characteristics of a transistor in CC configuration. **[7]**

Q.6 (a) Explain Zener Diode as a Voltage Regulator. **[7]**

(b) Explain the working of positive clamping circuit. **[7]**

Q.7 Explain the construction, working and operating characteristics of N-Channel JFET's with relevant diagrams **[14]**

Q.8 Describe the kind of operation that takes place in the enhancement mode MOSFET. How does this differ from depletion mode type? **[14]**

Q.9 Write Short notes on any two of the following **[7x2=14]**

(a) Characteristics of an Ideal Operational Amplifier
(b) Difference between Inverting and non inverting amplifier
(c) Effects of negative feedback in amplifiers
(d) Various topologies of feedback amplifier

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